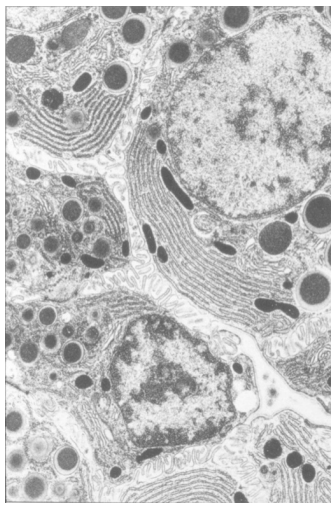
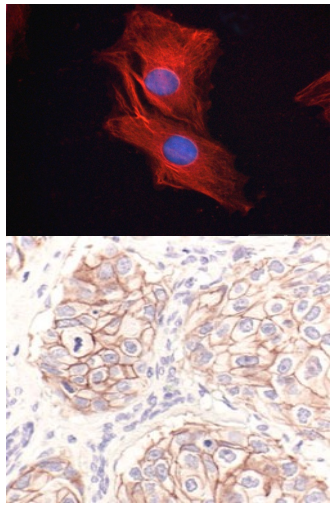
Light and electron microscopy

Lecture 1.2

September 1st, 2026



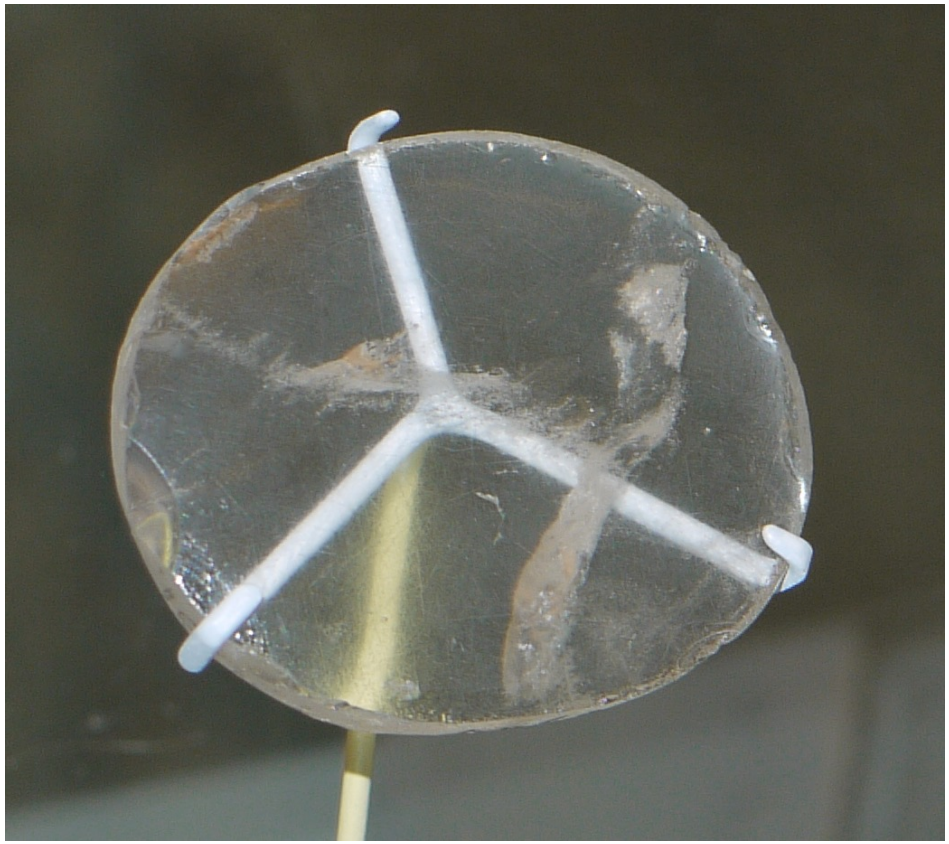
Dr. Serrano | dserrano@unav.es

1 Brief history of light microscopy

The Nimrud lens – year 750 BC

-Found in an arhaeological excavation in Iraq

-Unclear whether is a lens or an ornament



1 Brief history of light microscopy

Magnification lenses - 11th century

Cannot be attributed to any one person: Zacharias Janssen and Hans Lippershey, Johannes Kepler, Galilei…

Robert Hooke microscope.

Added a light source (oil lamp)
~1665
~ 50X magnification
Publishes Micrographia in 1667

→ **CELL BIOLOGY**





❷ Types of microscopes

OPTICAL	ELECTRONIC
CONVENTIONAL MICROSCOPE (light)	TRANSMISSION ELECTRONIC MICROSCOPE
PHASE CONTRAST MICROSCOPE	SCANNING ELECTRONIC MICROSCOPE
FLUORESCENCE MICROSCOPE	









② Light microscope

Microscope parts

Practical class

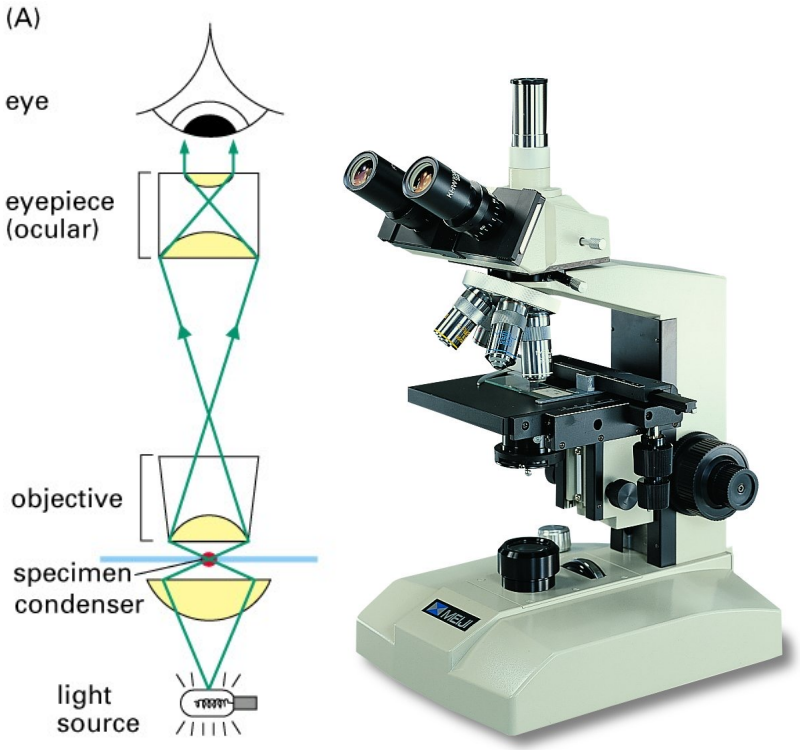
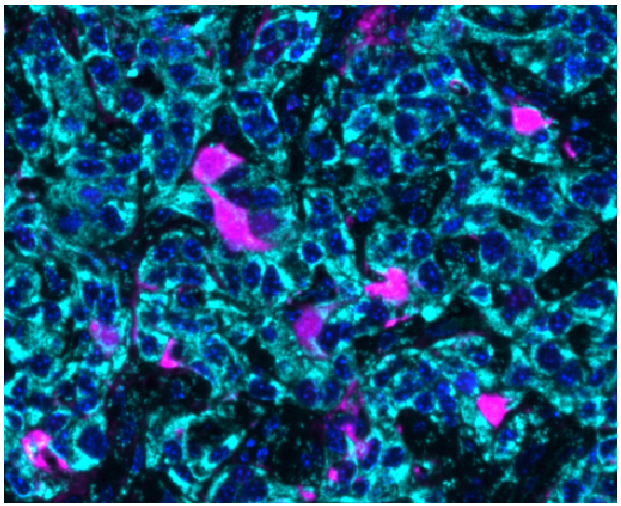
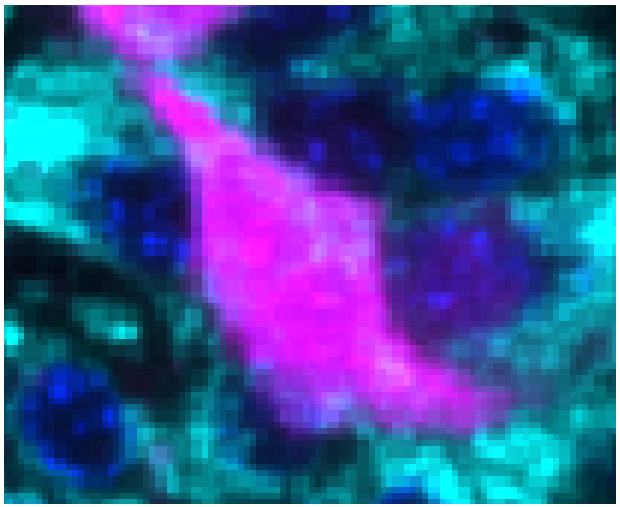


Figure 9–3. Molecular Biology of the Cell, 4th Edition.





-High magnification
-No sharpness

Low resolution

Magnification: Up to 1500 times

2.1 Resolution limit

Resolution limit

Ability to distinguish details in an image

Ernst Abbé (1873) defined resolution: *Minimum **distance** between two small points close to each other such that they can be resolved as two separate points*

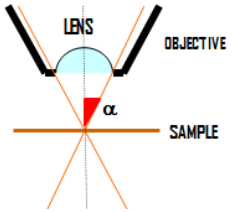
Resolution depends on:

-Quality of the sample

-Quality of the microscope (lenses)

- **λ** of light (wavelength)

$$LR(d) = \frac{0.61 \times \lambda}{n \times \sin \alpha}$$



LR(d) = Resolution limit. Lower is best
λ = wavelenght
n = refraction index
α = half-angle of maximun cone light entering the lens

Study the information about the formula to calculate “d” (resolution) in ADI

For the human eye: **d=0.1 mm**

For the regular light microscope: **d=0.22 µm**

Objects that are closer to each other or are smaller than 0.22 µm cannot be observed

2.1 It’s not only about resolution…

What can be seen with a light microscope?

-Mitochondria?

-Lysosomes?

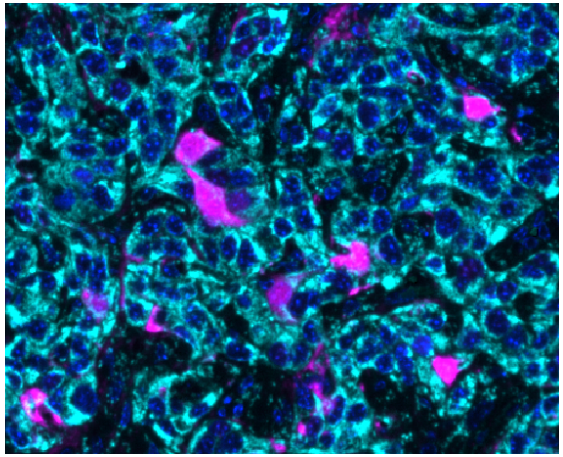
-Ribosomes?

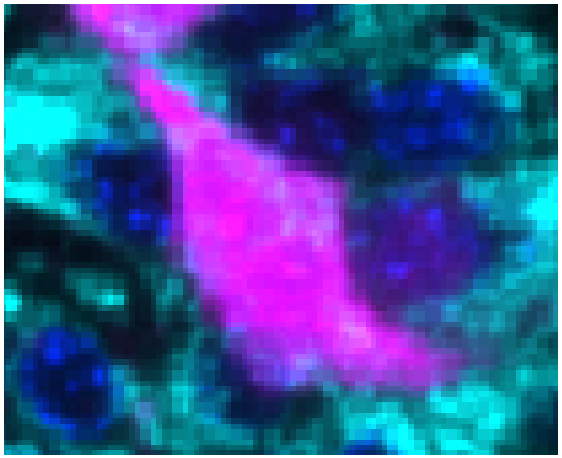
Two problems in cell biology

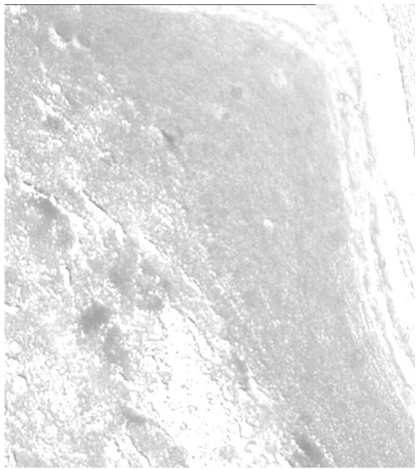
-Small size of cells and tissues ⇨ Microscope

-Translucency of tissues ⇨ ***Contrast:** -Differential staining

-Special optical systems







2.1 Contrast - staining

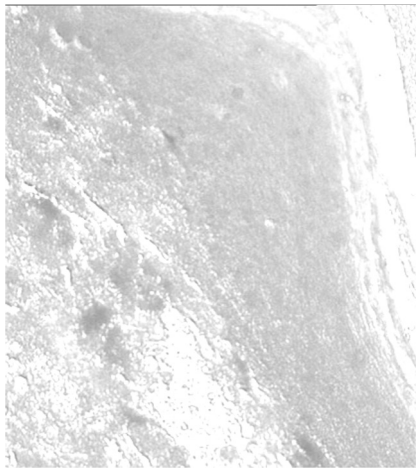
Creation of contrast by differential staining

(modification of λ)

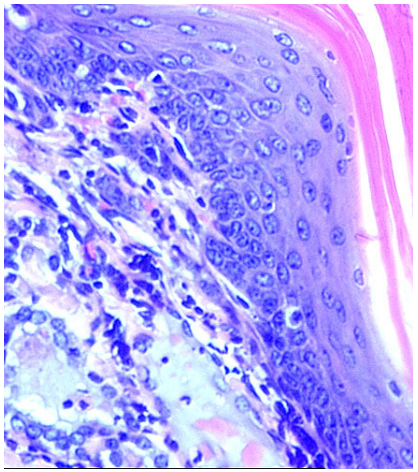


Spectrum of visible light

Sample without staining

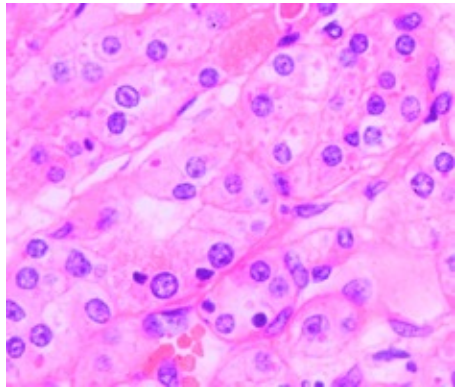


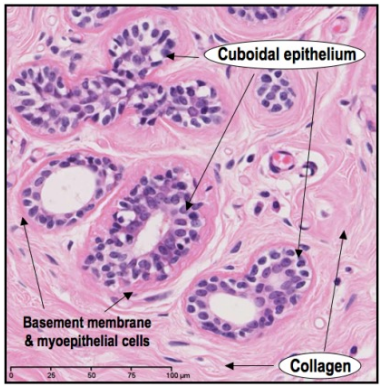
Sample with staining



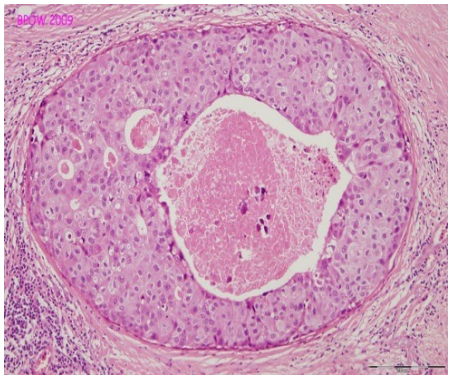
Haematoxylin-Eosin staining

1

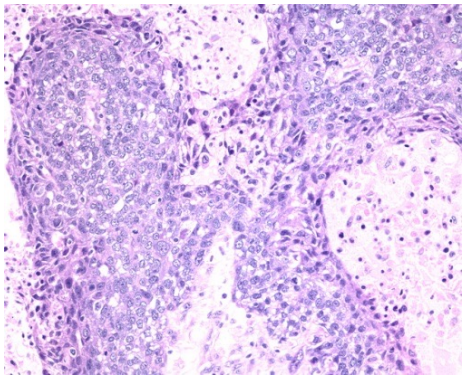




2



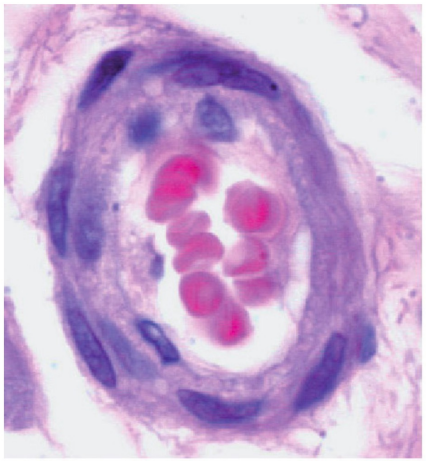
3

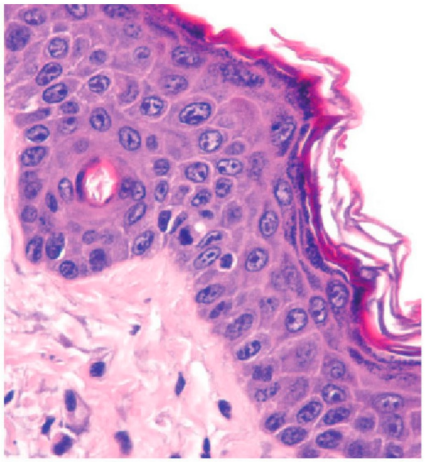


INFILTRATING TUMOR

2.1 Contrast - staining

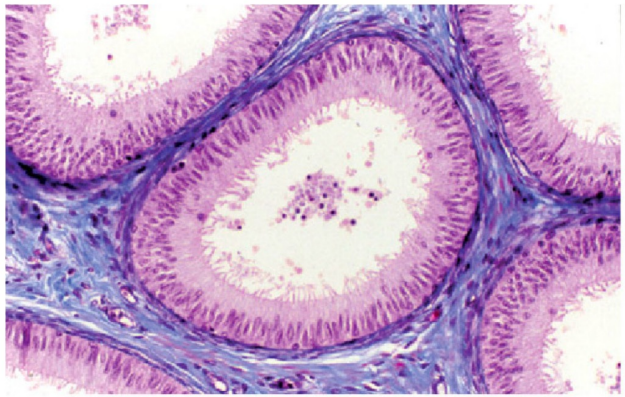
Haematoxylin-Eosin staining

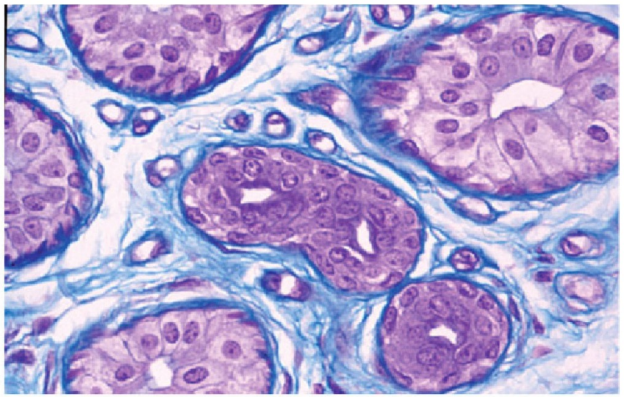




Nuclear staining: **hemoatoxylin**
Cytoplasm stain: **eosin**

Masson trichrome staining



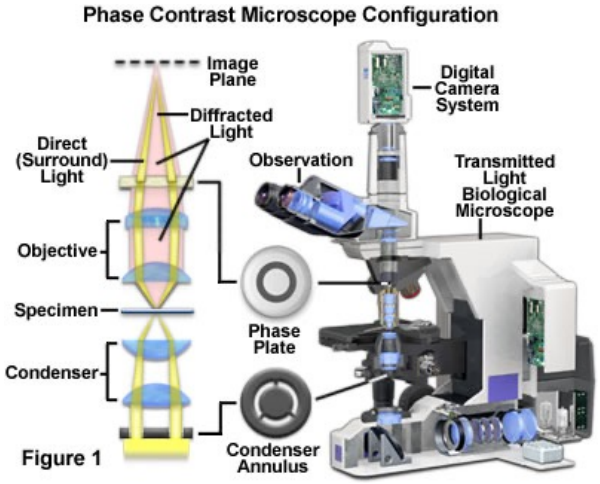


Nuclear staining: **hemoatoxylin**
Fiber staining (collagen): **aniline blue**
Cytoplasm stain: **Ponceau G**

2.1 Contrast without staining

Special optical systems

PHASE CONTRAST MICROSCOPE



Based on phase (wave) shifts when light passess through different objects

Phase shifts are invisible to the naked eye

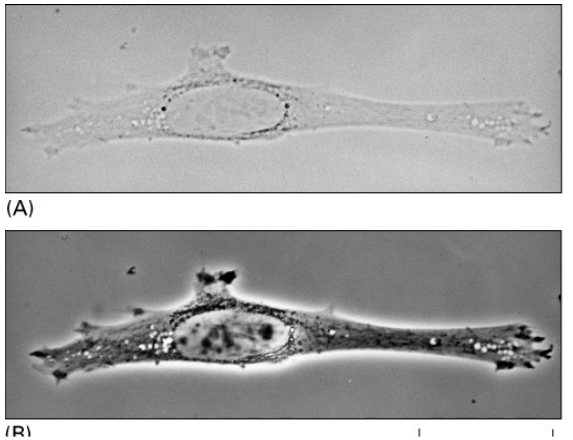
Phase contrast microscope alters light pathways:

-Background

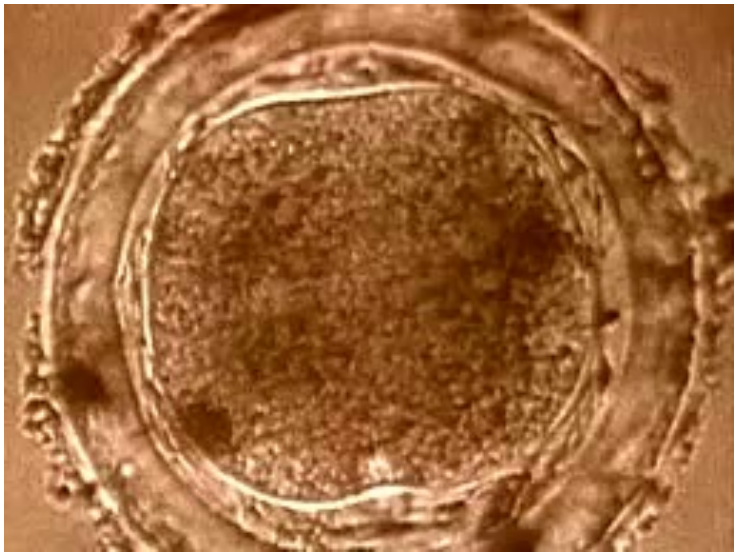
-Scattered

→ **CONTRAST**

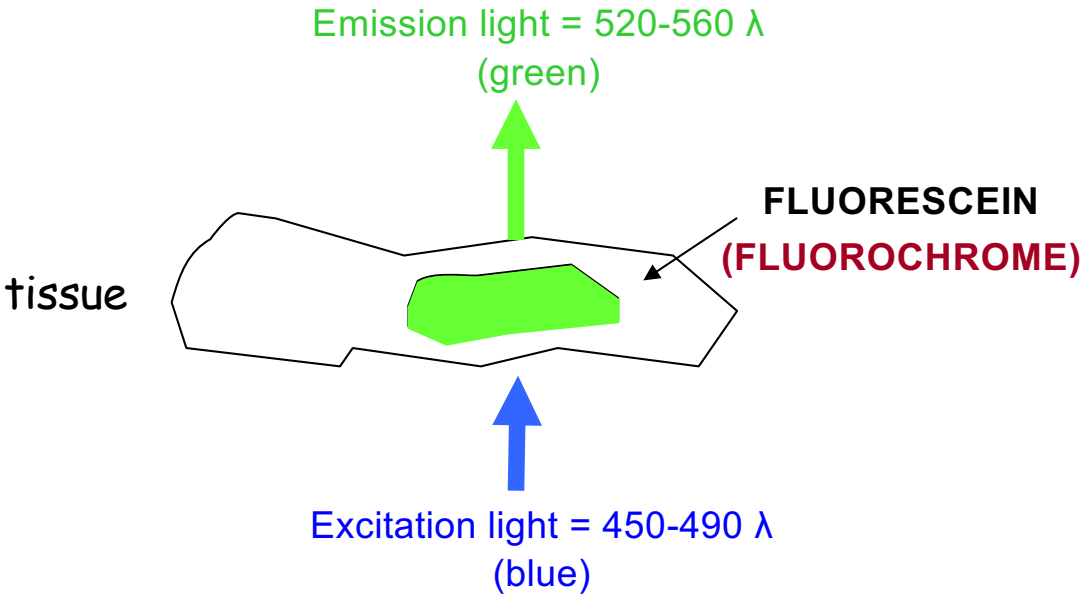
Regular light microscope

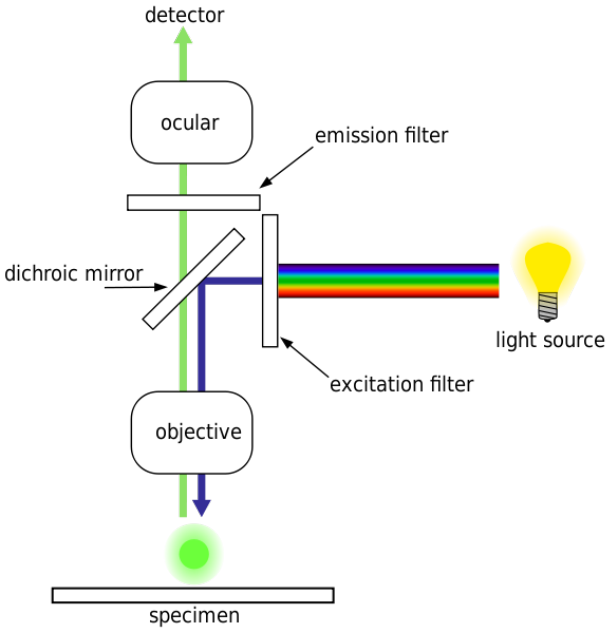


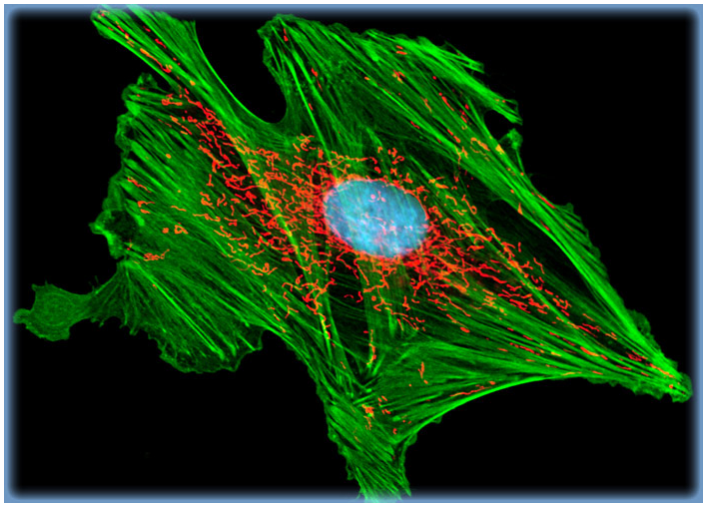
Phase contrast

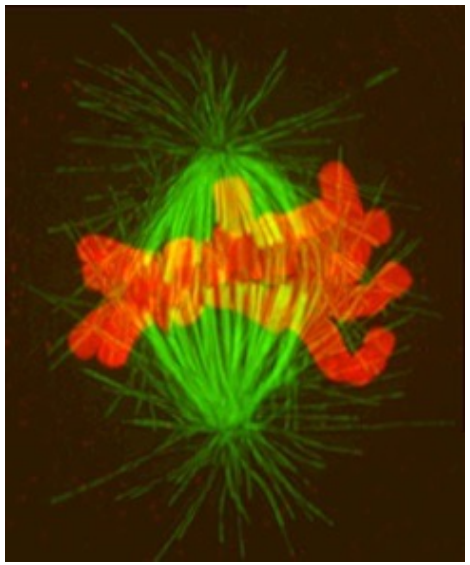


3 Fluorescence microscope









How to process samples for their observation by light microscopy

(in Spanish)

https://www.youtube.com/watch?v=KyO3TT2NaKk

https://www.youtube.com/watch?v=EMXpQzwOc_Q

4 Electron microscopes





4.1 Transmission electron microscope (TEM)

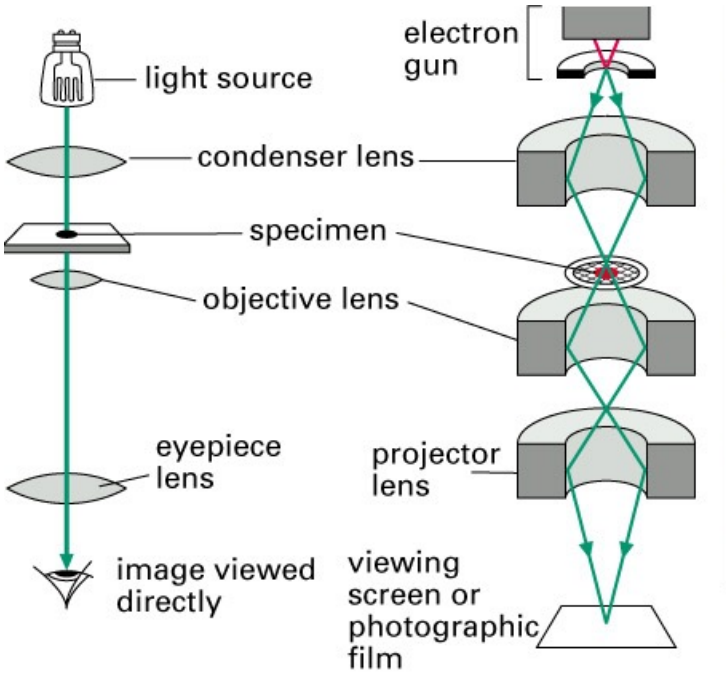
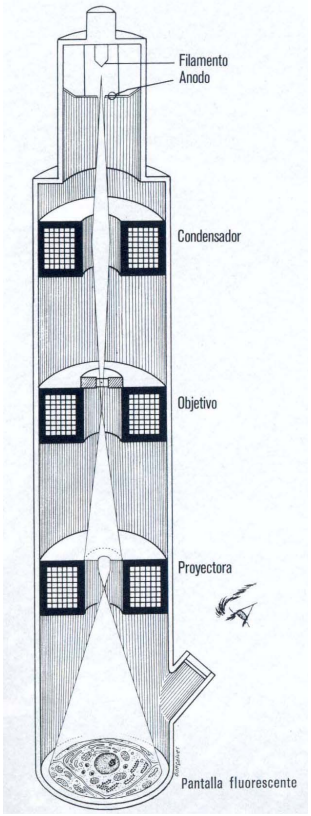


Figure 9–22. Molecular Biology of the Cell, 4th Edition.





-Beam of electrons

-Electromagnetic coil

-High vacuum

-Magnification: 150,000

-Limit of resolution: 10-20 A

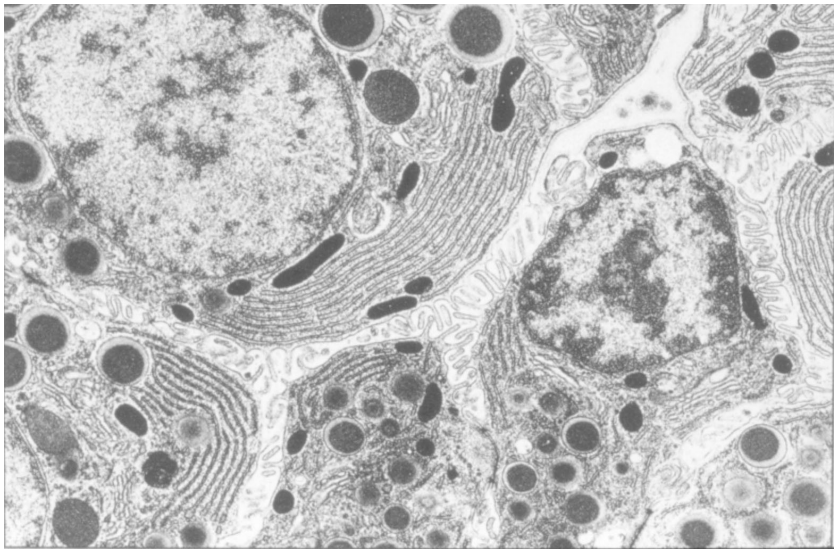
4.1 Transmission electron microscope (TEM)

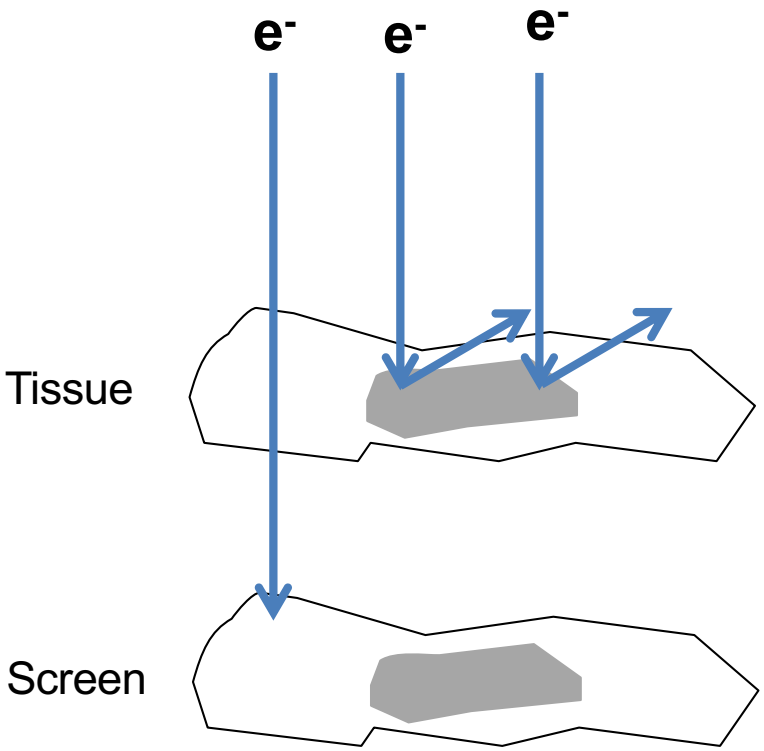
-Ultra-thin sections (0.05 µm)

-Contrast:

-Uranium acetate

-Lead citrate





Contrast: Differential absorption of heavy metal salts

Dark areas: **Electron-dense**

Bright areas: **Electron-lucent**

4.1 TEM – Organelle visualization

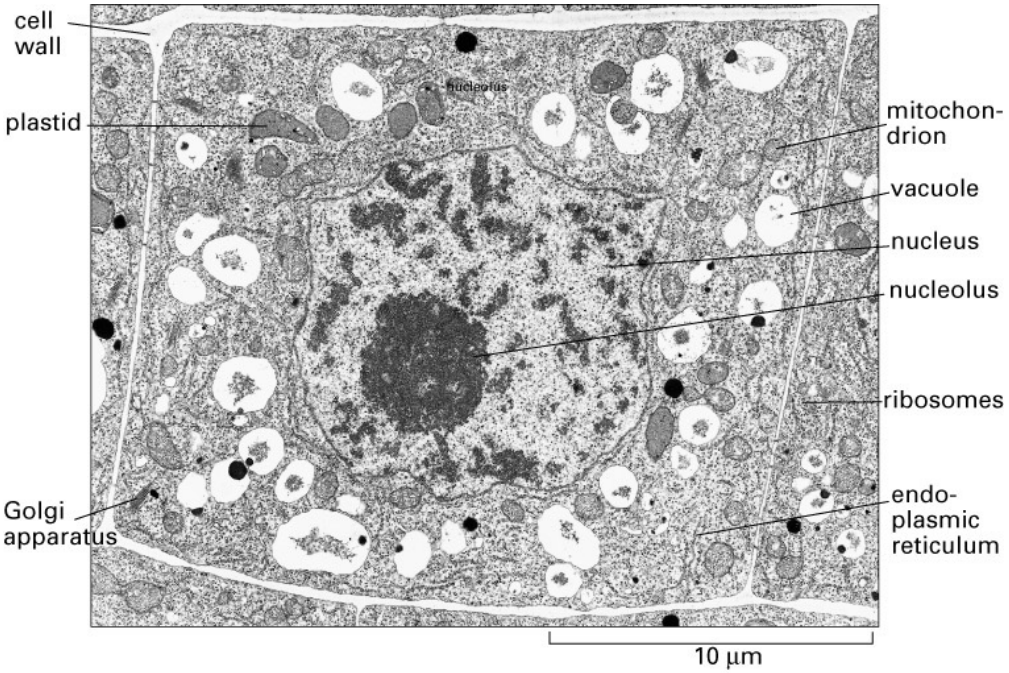


Figure 9–25. Molecular Biology of the Cell, 4th Edition.



Some symptoms:

-Infertility

-Difficulties getting pregnant

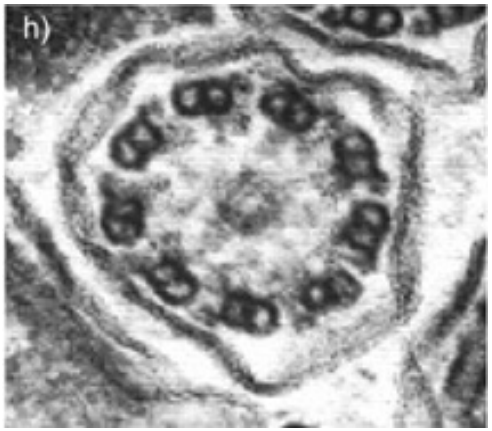
-Deafness

-Bronchitis and recurrent infections

Normal cilium



Dyskinesia



4.1 Scanning electron microscope (SEM)

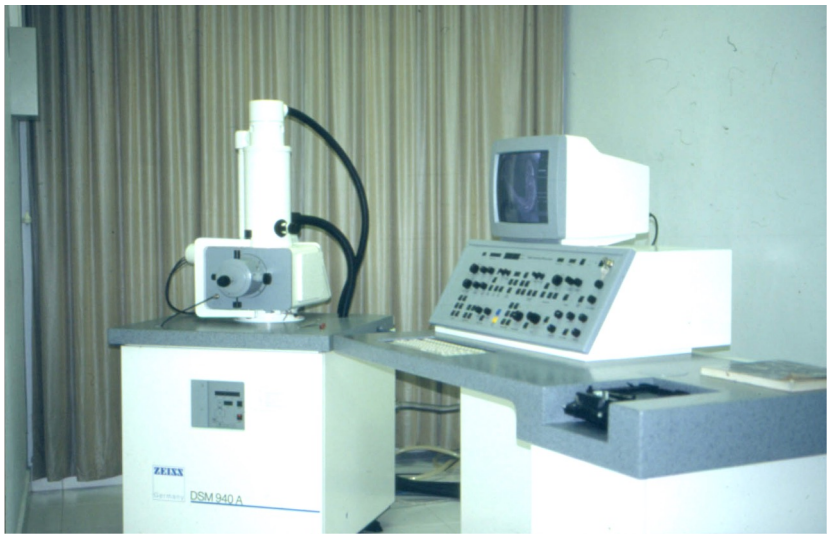
-Three-dimensional structures (cell and tissue surfaces)

-Gold coating

-The electron beam scans the sample

-Observation on a TV monitor

-30000X magnification



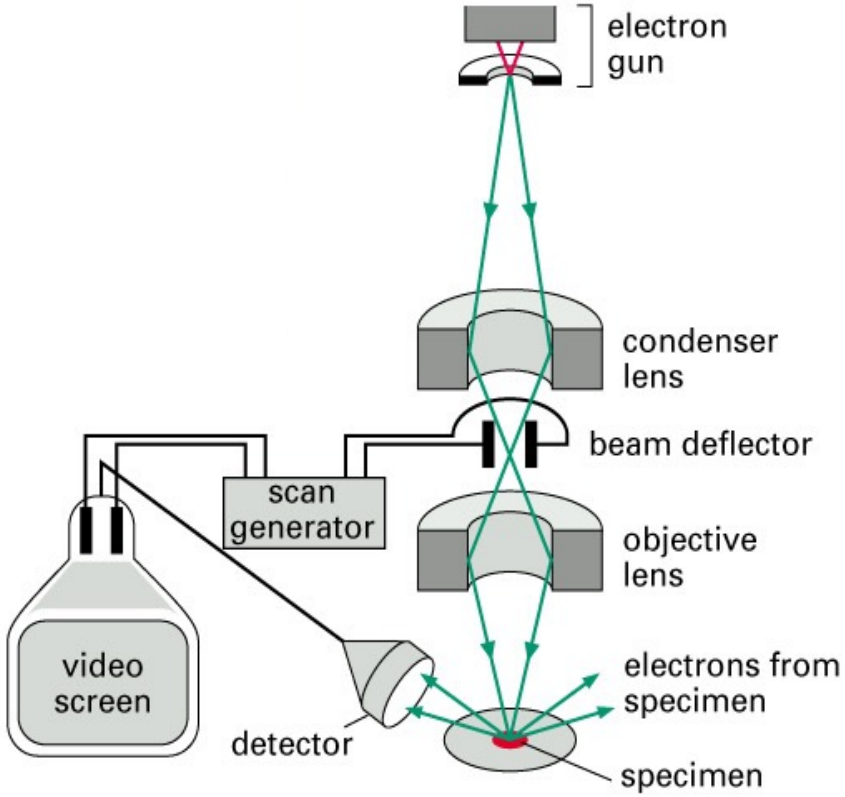
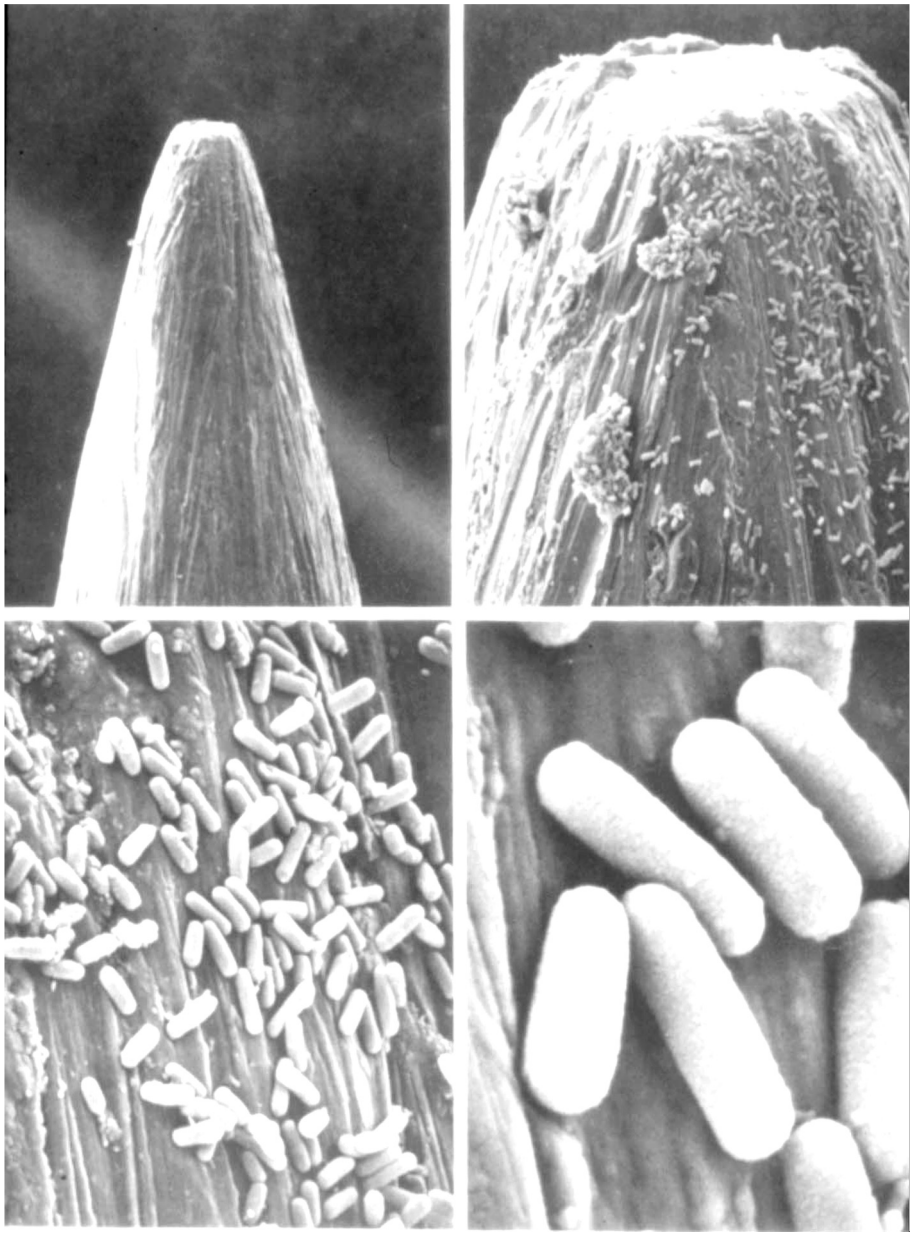
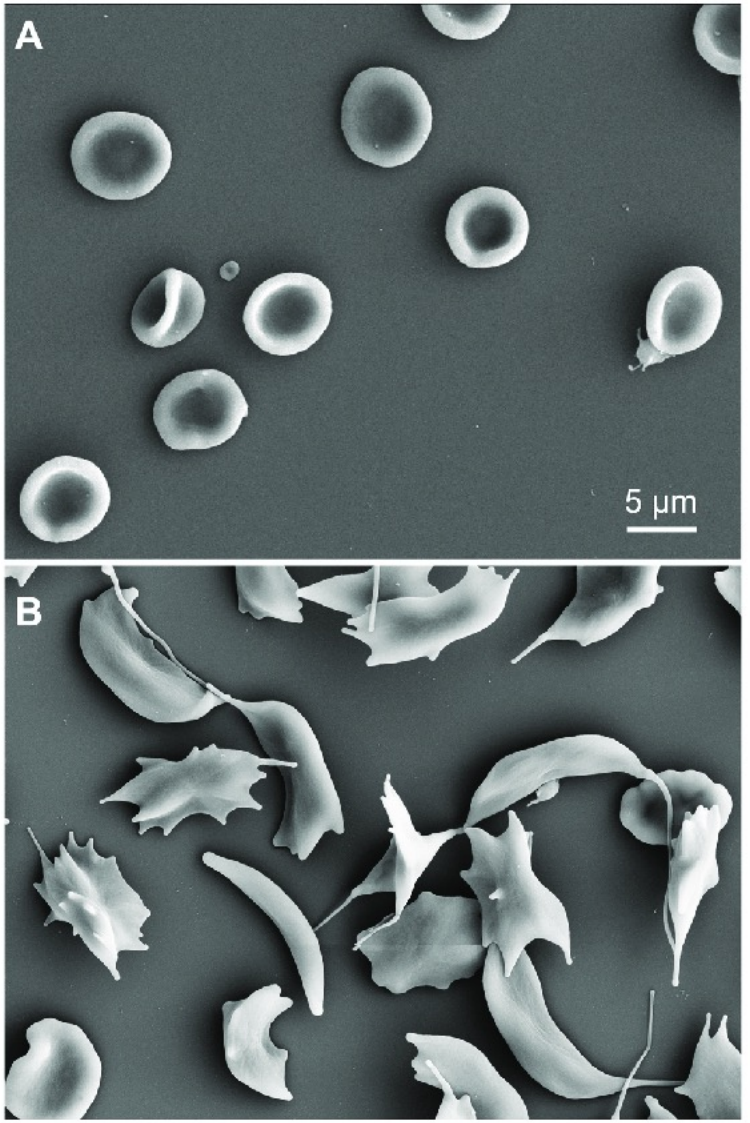


Figure 9–29. Molecular Biology of the Cell, 4th Edition.

4.1 Scanning electron microscope (SEM)





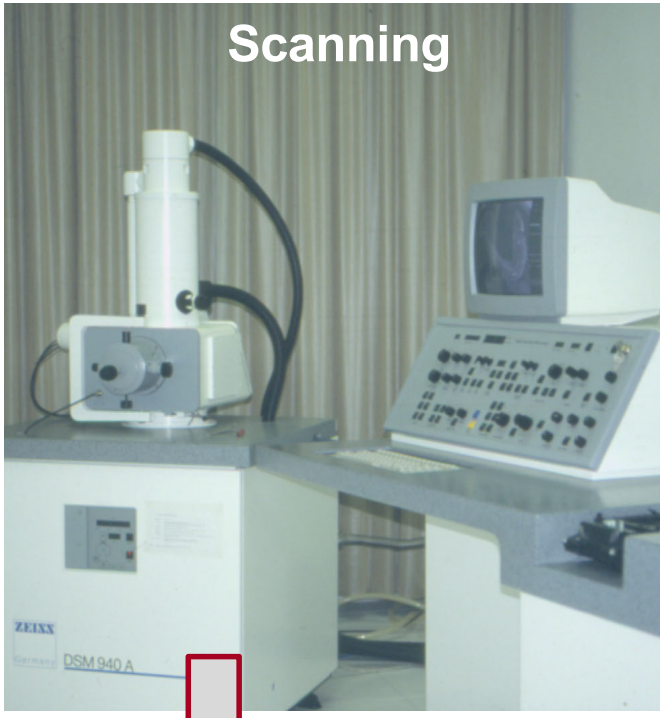
Sicle cell disease

4 Electron microscopes

Electron microscopes at the University







Dual: Transmission + Scanning

The principles of electron microscope.

Uploaded to adi

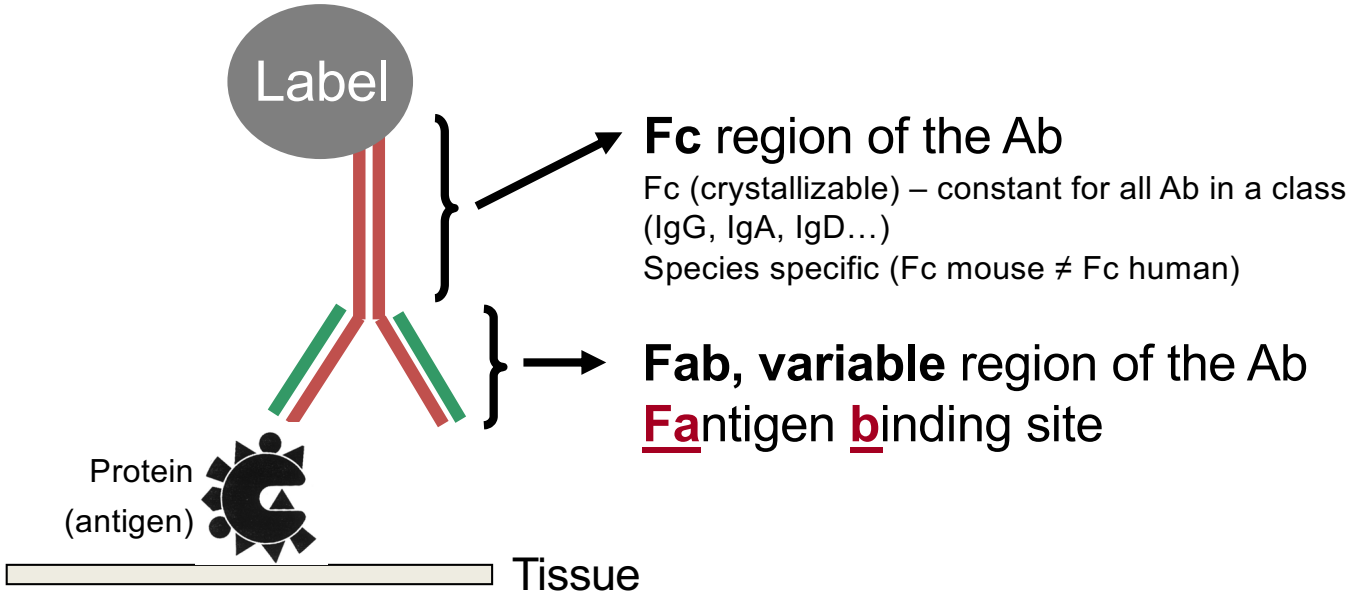
The electron microscope. How to process samples electron microscopy

(in Spanish)

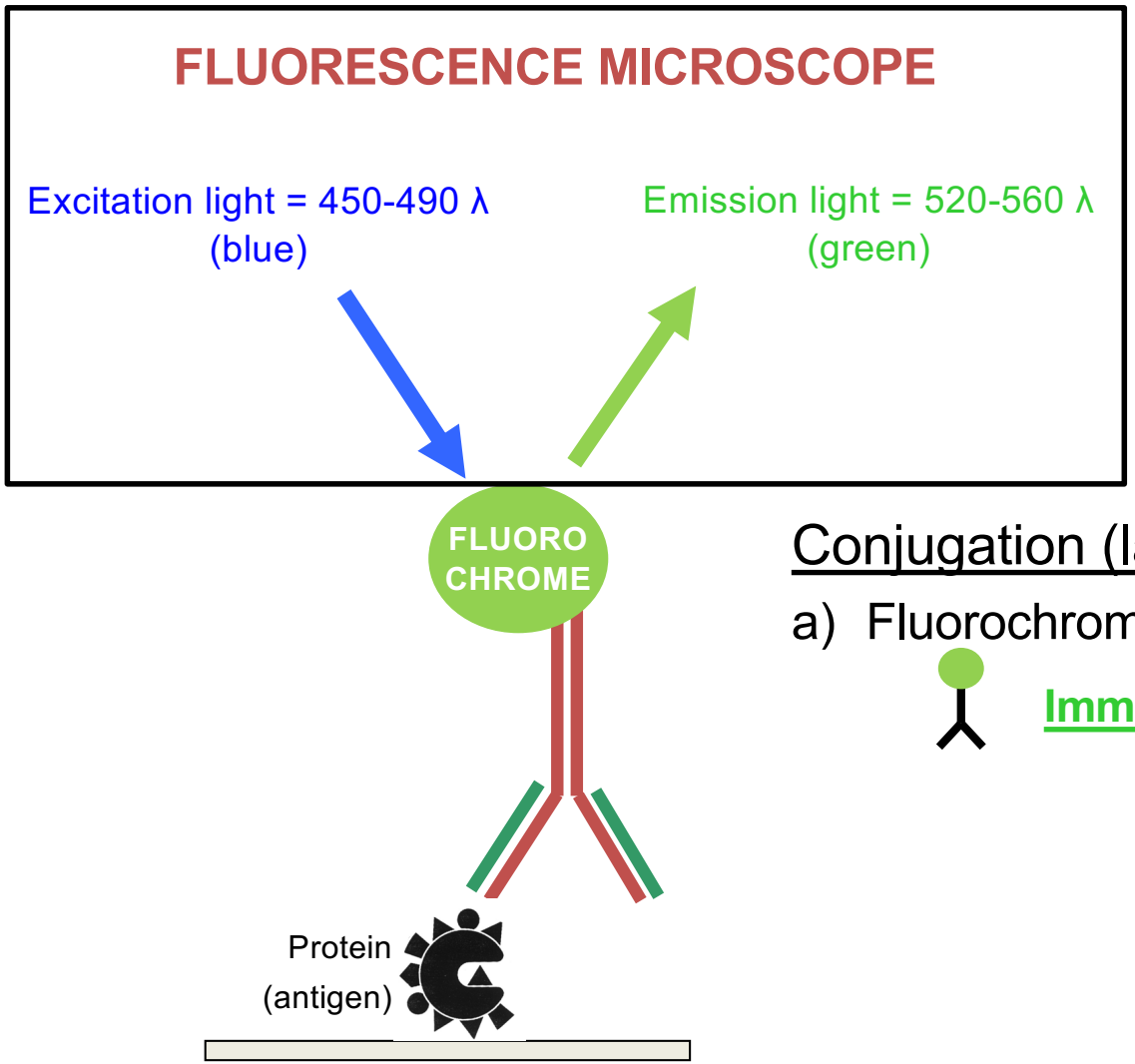
https://www.youtube.com/watch?v=TfbOez7gHXA

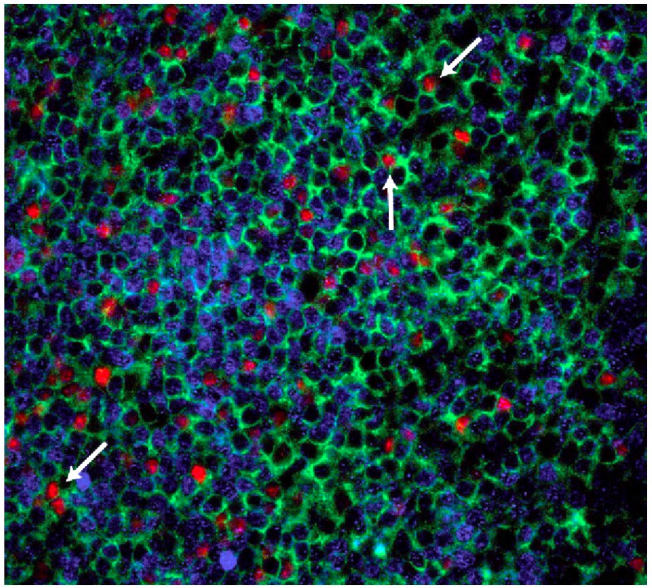
5 Applications in biology - immunodetection

Antibody (Ab) (immunoglobulin)



5 Applications in biology - immunodetection





Conjugation (label):

a) Fluorochromes: fluorescein, rhodamin…

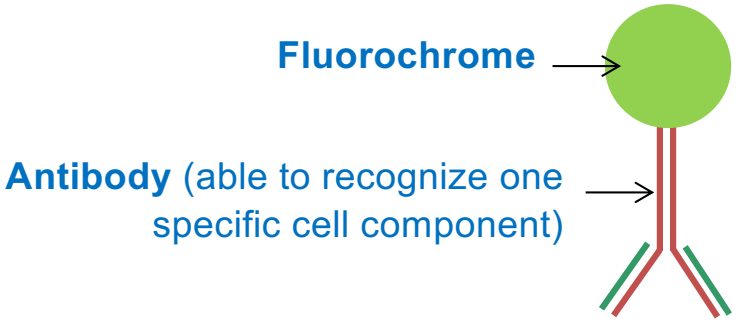
Immunofluorescence (IF)

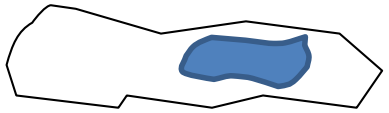
5.1 Immunodetection in fluorescence microscopy

-In combination with **antibodies** (**immunofluorescence IF**): Identify specific parts/organelles of the cell

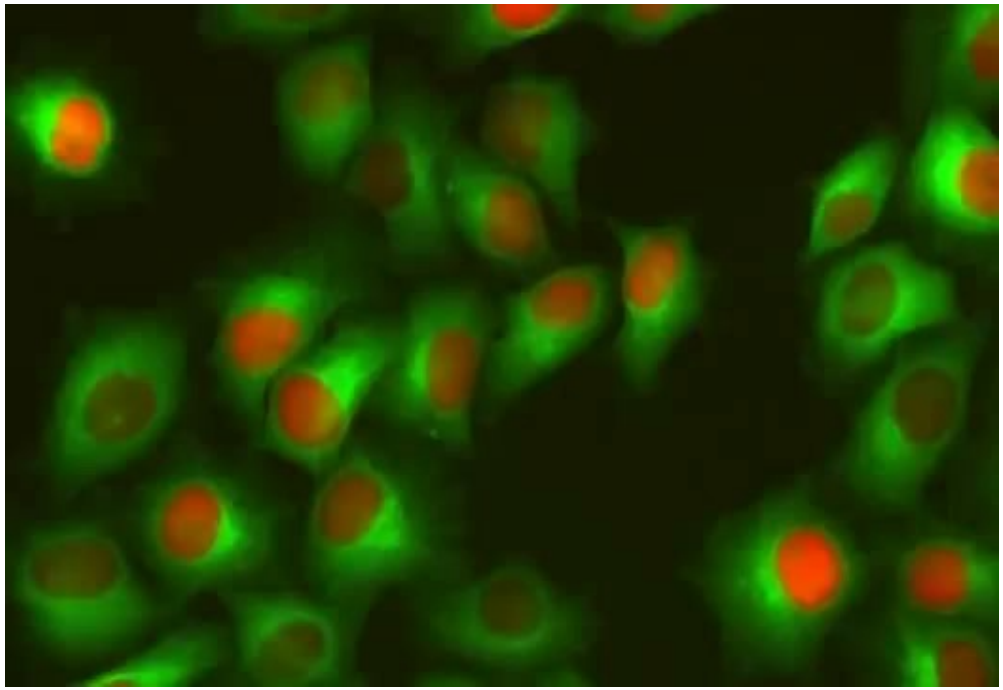
-*In vivo* studies

-Metastasis models





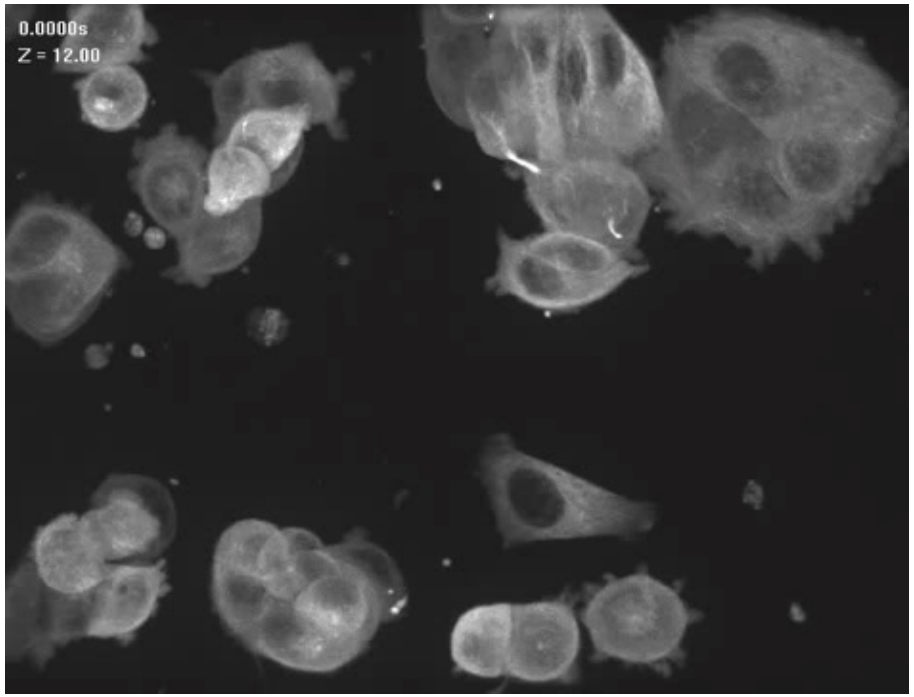
Specimen



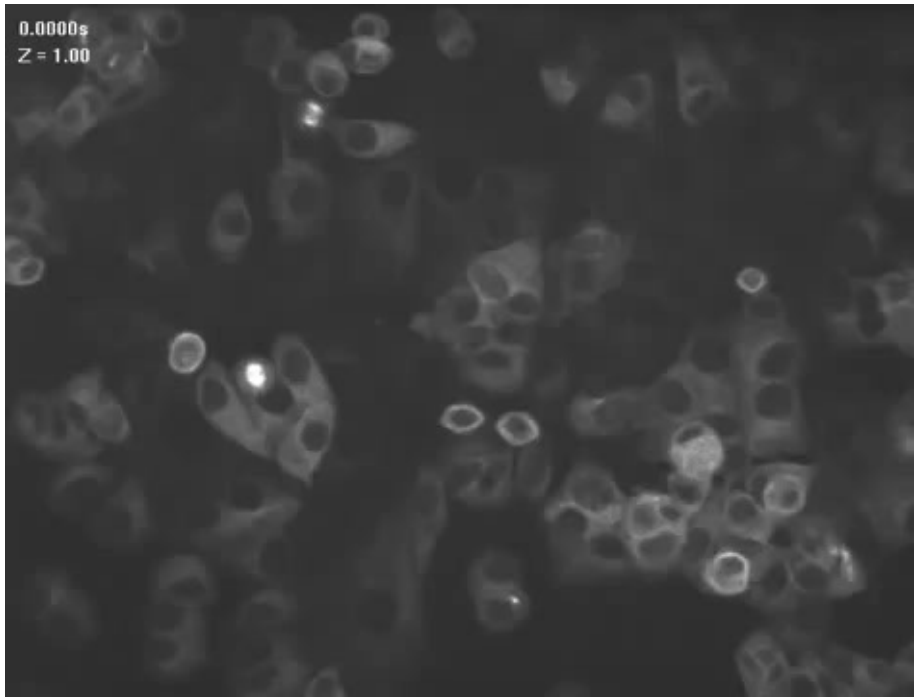
5.1 Importance of IF in cancer research

Development of new drugs

Breast cancer cells in culture



Breast cancer cells in culture
treated with Docetaxel
(inhibitor of microtubules)

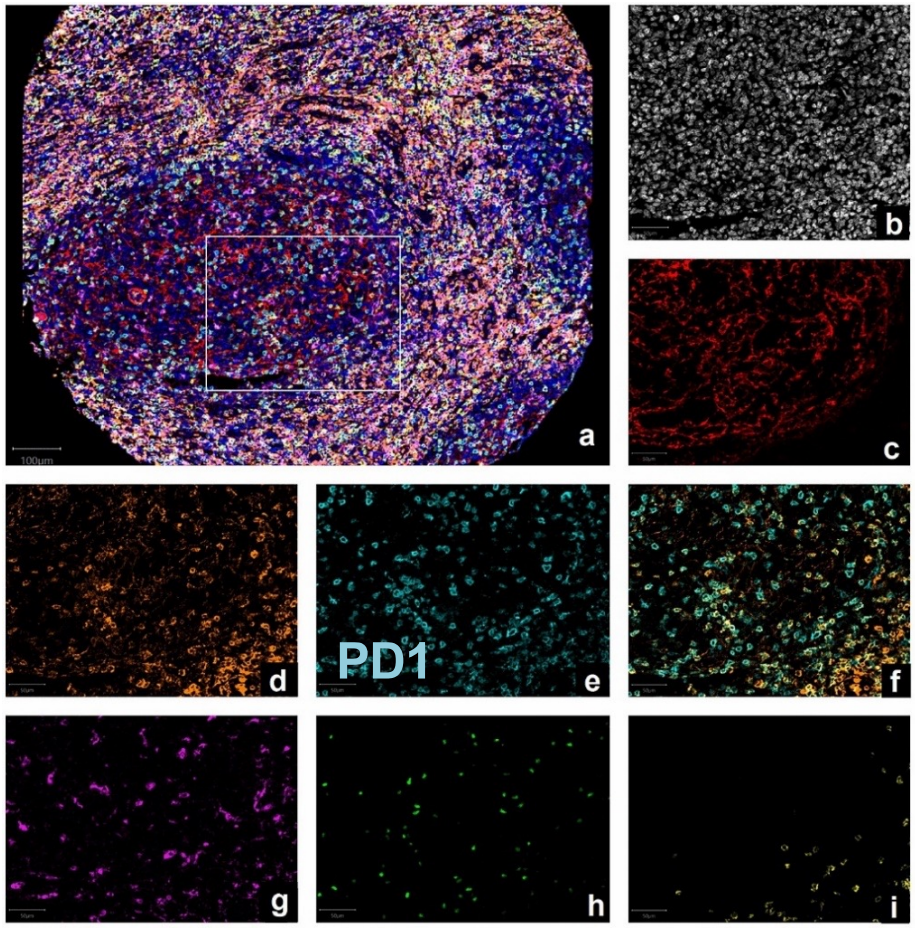


5.1 Importance of IF in cancer research

Multiplex IF in lymphoma

Multiplexed IF analysis: allows the simultaneous study of markers of malignancy and immune response in the same sample.

Personalized medicine



DAPI
DNA stain

CD21
Follicular cell

CD4
T helper cell

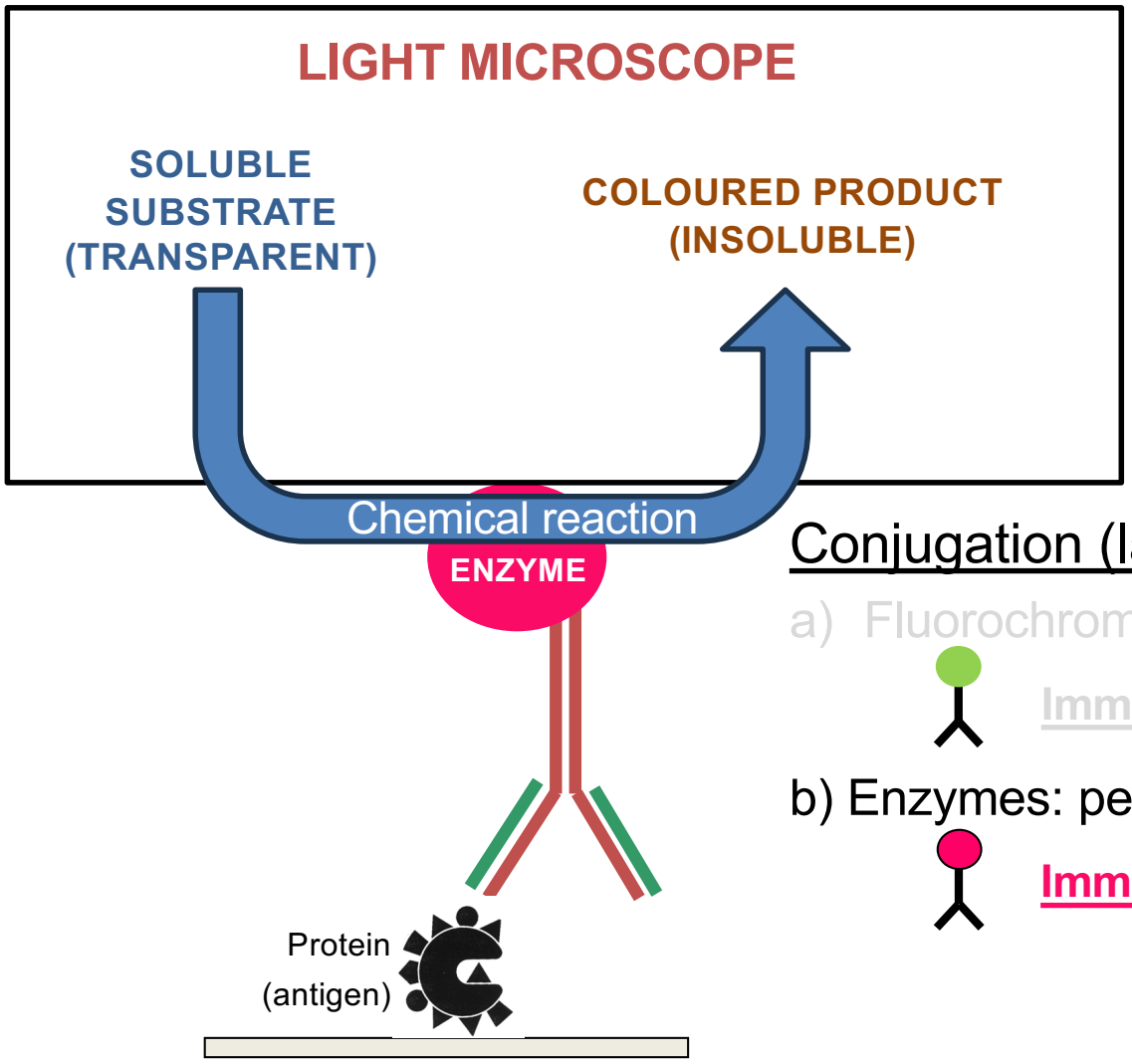
CD4 + PD1
PD1: PD-L1 Receptor. Regulates immune response

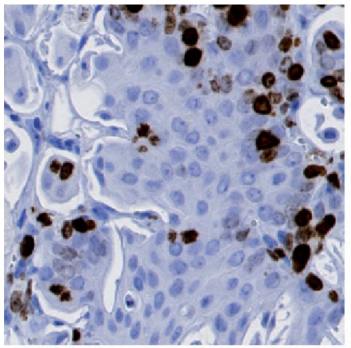
CD68
Macrophages

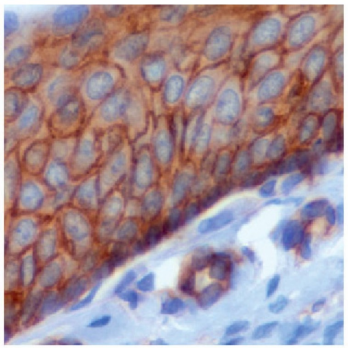
CD8
Cytotoxic T cell

FOXP3
T regulatory cell

5 Applications in biology - immunodetection







Conjugation (label):

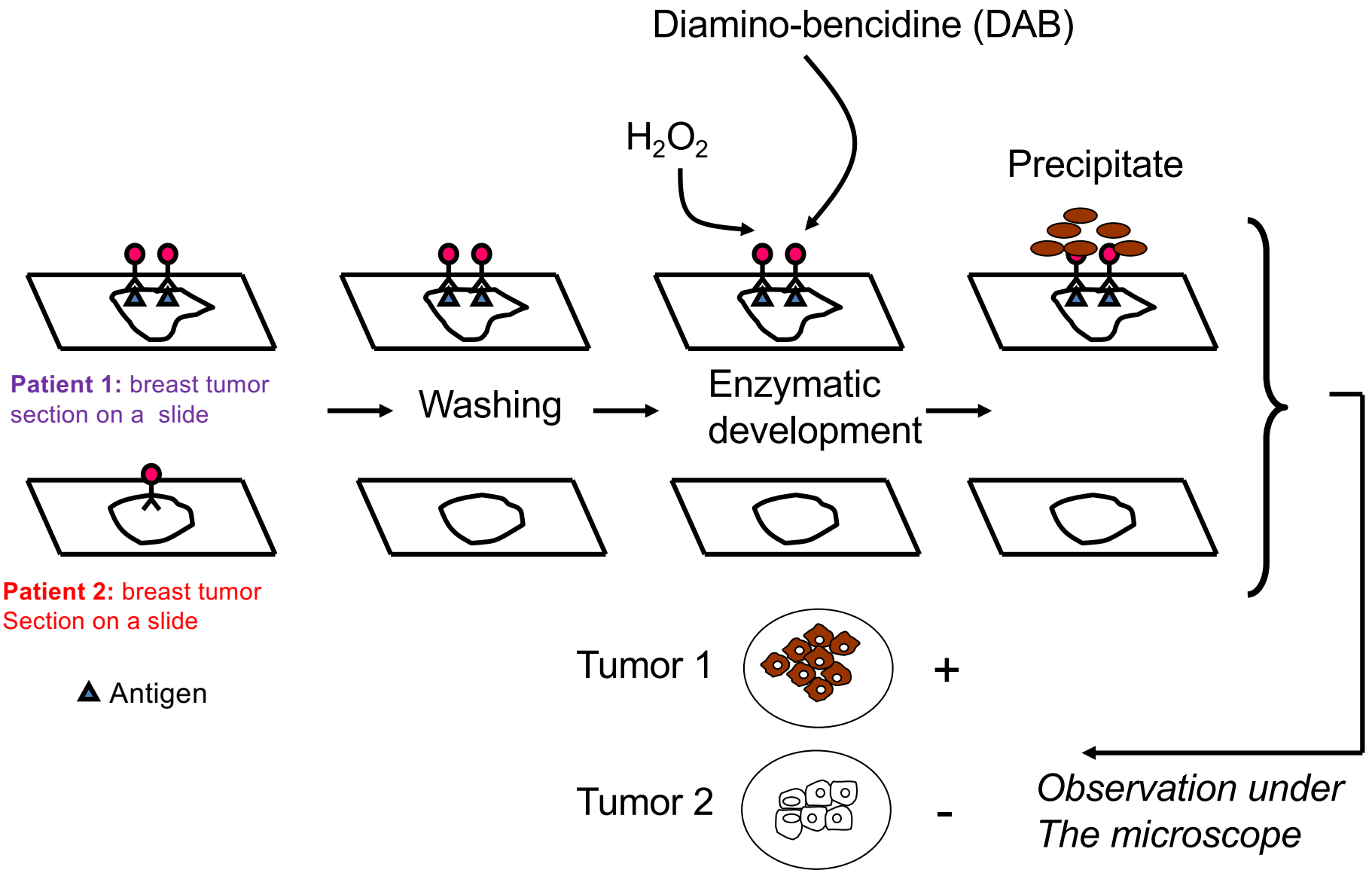
a) Fluorochromes: fluorescein, rhodamin…

Immunofluorescence (IF)

b) Enzymes: peroxidase, alkaline phosphatase

Immunohistochemistry (IHC)

5.2 Immunohistochemical (IHC) Protolocol

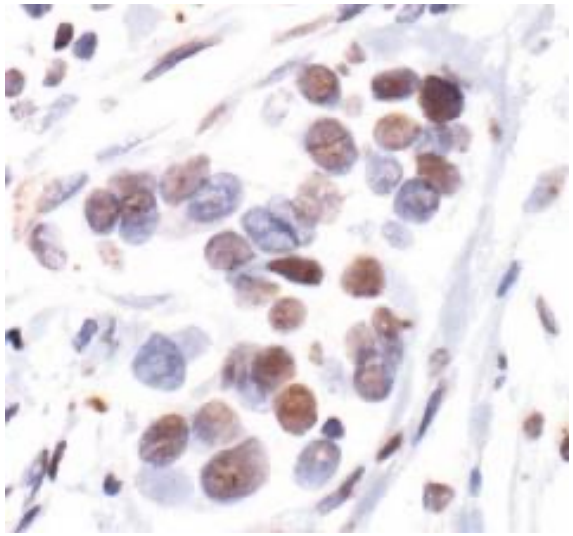


5.2 Use of IHC in clinical practice

Molecular markers → Diagnosis → Treatment

Molecular markers → Prognosis

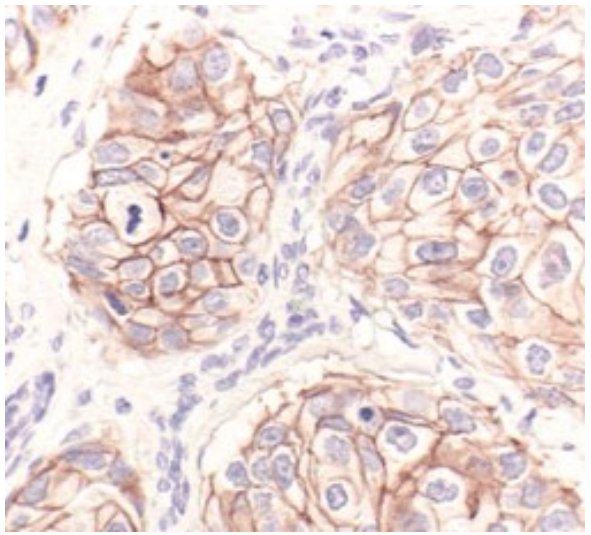
Example: Breast cancer



Estrogen receptor (ER)

↓

Treatment with anti-estrogens (tamoxifen)

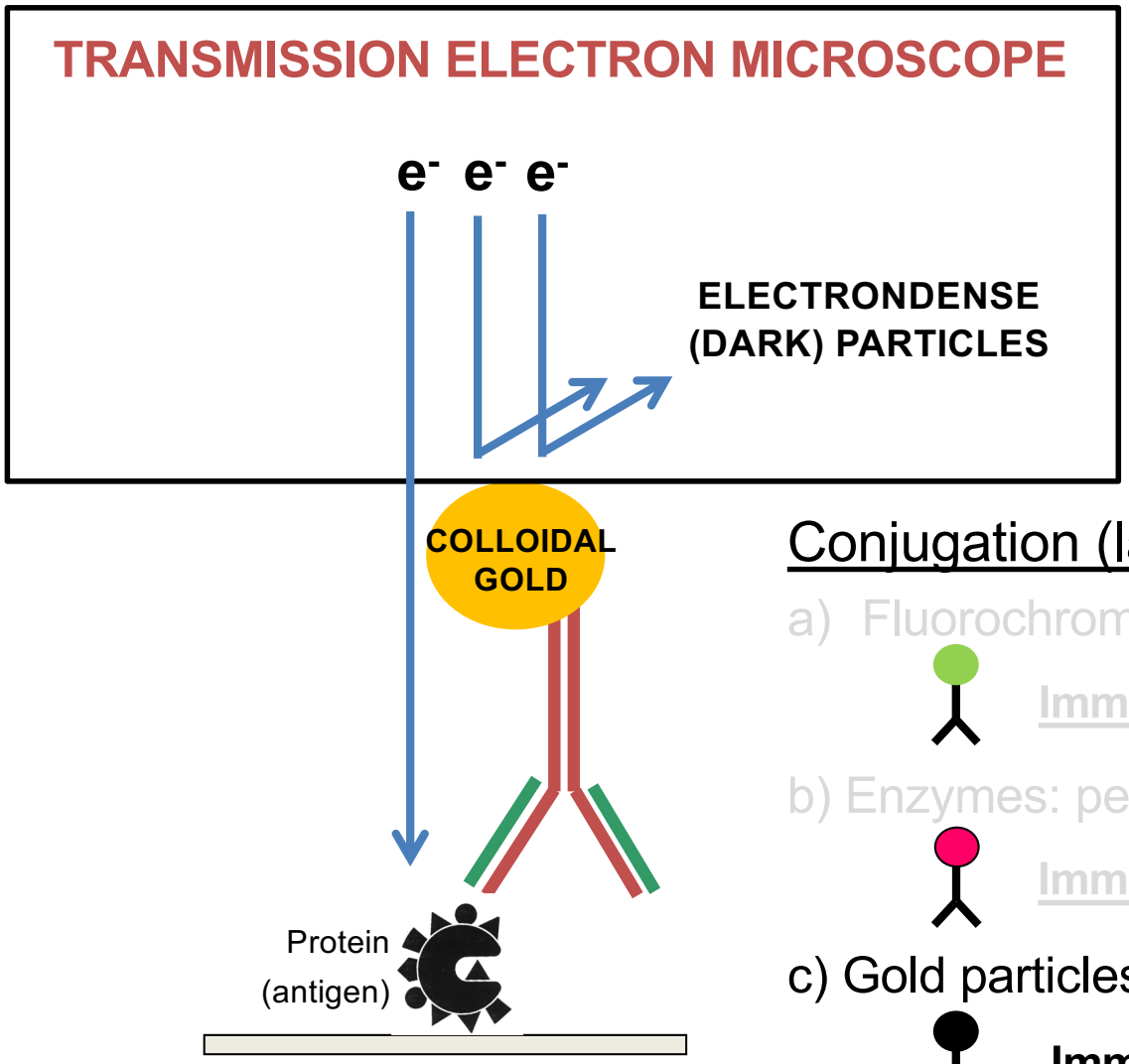


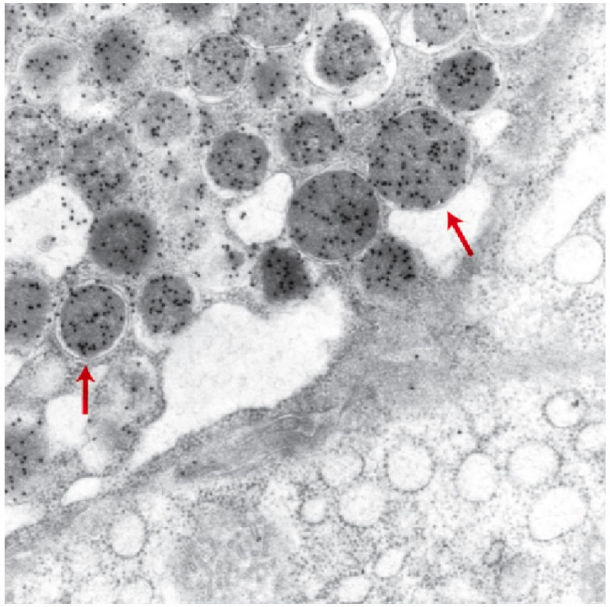
Her-2/Erbb2

↓

Treatment with Transtuzumab (Herceptin)

5 Applications in biology - immunodetection





Conjugation (label):

a) Fluorochromes: fluorescein, rhodamin…

Immunofluorescence (IF)

b) Enzymes: peroxidase, alkaline phosphatase

Immunohistochemistry (IHC)

c) Gold particles



Immunoelectron microscopy